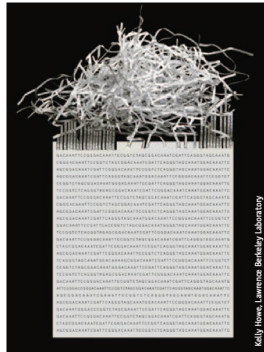


Genome Assembly



Assemblers need copious sequencing data and informatic exertion to put the genome back together.

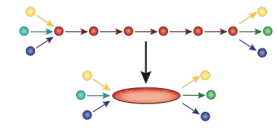
1. Fragment DNA and sequence



2. Find overlaps between reads

...AGCCTAGACCTACA**GGATGCGGACACGT**
GGATGCGGACACGTCGCATATCCGGT...

3. Assemble overlaps into contigs



4. Assemble contigs into scaffolds



Genome assembly stitches together a genome from short sequenced pieces of DNA.

Some slides throughout the next sections are credited to Chris Mason, Ben Langmead, and Mike Schatz
 Baker, M. Nature Methods, 2012

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Genome Assembly

Reads



+

Reference genome



Input DNA



How to assemble puzzle without the benefit of knowing what the finished product looks like?

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Assembly

Whole-genome “shotgun” sequencing starts by copying and fragmenting the DNA

(“Shotgun” refers to the random fragmentation of the whole genome; like it was fired from a shotgun)

Input: GCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT

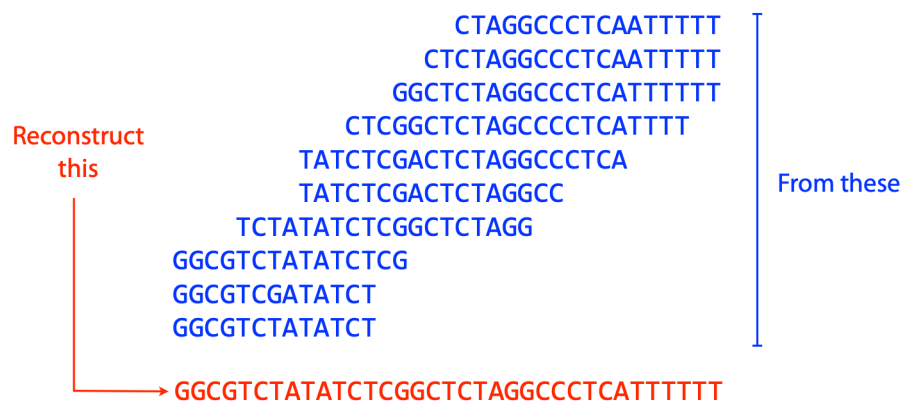
Copy: GCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT
 GCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT
 GCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT
 GCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT

Fragment: GCGTCTA TATCTCGG CTCTAGGCCCTC ATTTTTT
 GGC GTCTATAT CTCGGCTCTAGGCCCTCA TTTTTT
 GCGTC TATATCT CGGCTCTAGGCCCT CATTTTTT
 GCGTCTAT ATCTCGGCTCTAG GCCCTCA TTTTTT

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Assembly

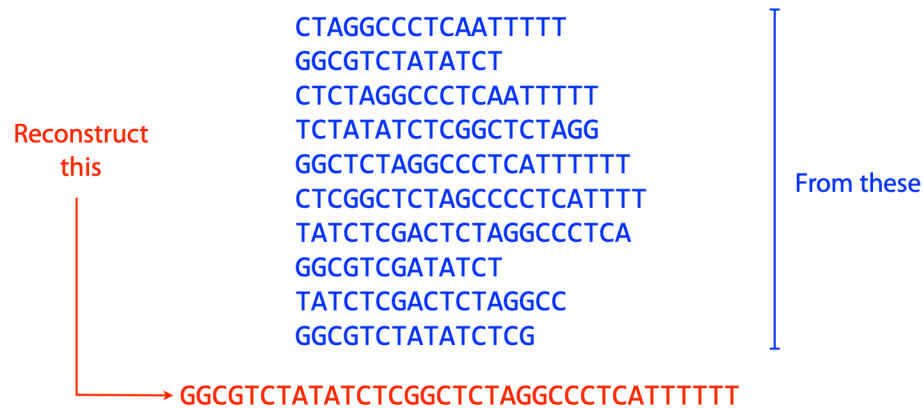
Assume sequencing produces such a large # fragments that almost all genome positions are covered by many fragments...



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Assembly

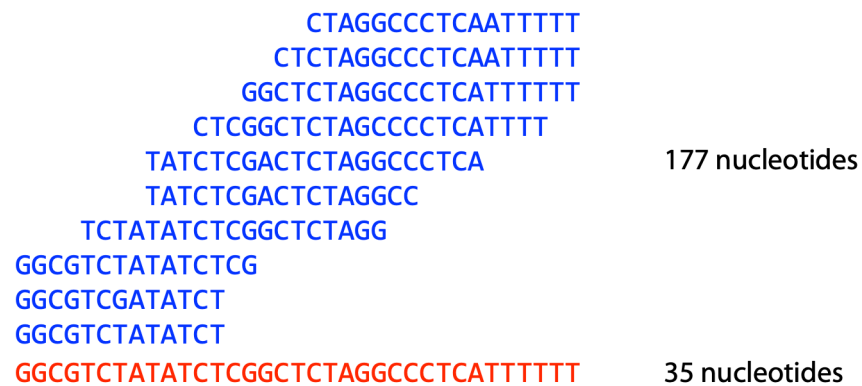
...but we don't know what came from where



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Assembly

Key term: *coverage*. Usually it's short for *average coverage*: the average number of reads covering a position in the genome.

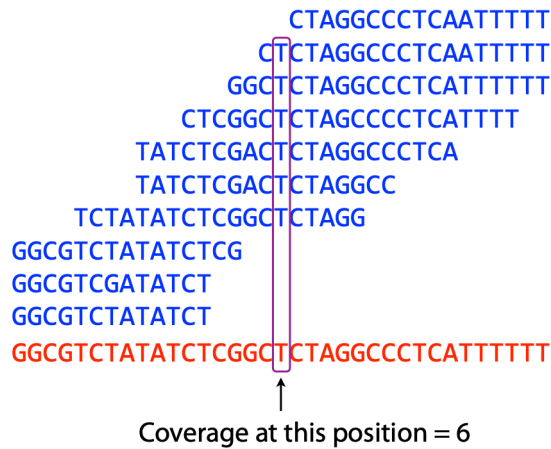


$$\text{Average coverage} = 177 / 35 \approx 5x$$

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Assembly

Coverage could also refer to the number of reads covering a particular position in the genome:



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First law of assembly

If a suffix of read A is similar to a prefix of read B...

TCTATATCTCGGCTCTAGG
 ||||| |||||
 TATCTCGACTCTAGGCC

...then A and B might *overlap* in the genome

TCTATATCTCGGCTCTAGG
 GCGTCTATATCTCGGCTCTAGGCCCTCAATTTT
 TATCTCGACTCTAGGCC

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Assembly

Say two reads truly originate from overlapping stretches of the genome. Why might there be differences?

```

      TATCTCGACTCTAGGCC
      ||||| |||||
TCTATATCTCGGCTCTAGG
      ↑
  
```

1. Sequencing error
2. Difference between inherited *copies* of a chromosome
E.g. humans are diploid; we have two copies of each chromosome, one from mother, one from father. The copies can differ:

Read from Mother: TATCTCGACTCTAGGCC

Read from Father: TCTATATCTCGGCTCTAGG

Sequence from Mother: TCTATATCTCGACTCTAGGCC

Sequence from Father: TCTATATCTCGGCTCTAGGCC

We'll mostly ignore ploidy, but real tools must consider it

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Second law of assembly

More coverage leads to more and longer overlaps

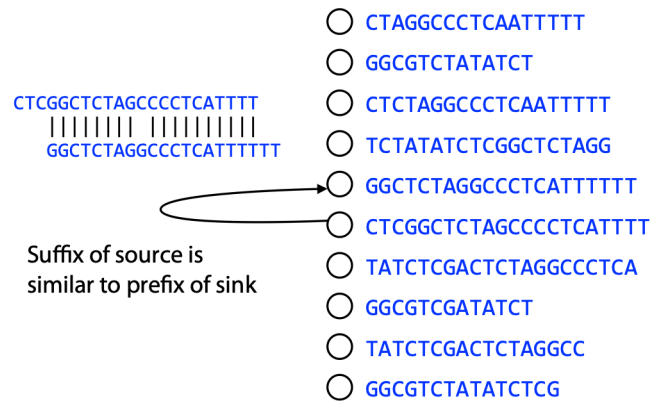
```

                        CTAGGCCCTCAATTTT
                CTCGGCTCTAGGCCCTCATTTT
      TCTATATCTCGGCTCTAGG      less coverage
GGCGTCGATATCT
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTT
                        CTAGGCCCTCAATTTT
                        GGCTCTAGGCCCTCATTTT
                        CTCGGCTCTAGGCCCTCATTTT
                TATCTCGACTCTAGGCCCTCA
      TCTATATCTCGGCTCTAGG
GGCGTCTATATCTCG
GGCGTCTATATCT      more coverage
  
```

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Overlaps

Finding all overlaps is like building a *directed graph* where directed edges connect overlapping nodes (reads)



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Directed graph review

Directed graph $G(V, E)$ consists of set of *vertices*, V and set of *directed edges*, E

Directed edge is an *ordered pair* of vertices.
First is the *source*, second is the *sink*.

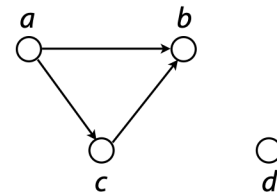
Vertex is drawn as a circle

Edge is drawn as a line with an arrow connecting two circles

Vertex also called *node* or *point*

Edge also called *arc* or *line*

Directed graph also called *digraph*



$V = \{a, b, c, d\}$
 $E = \{(a, b), (a, c), (c, b)\}$

↑
↑
 Source Sink

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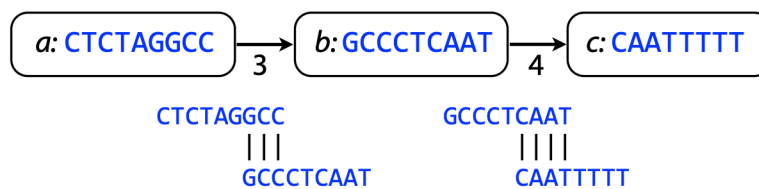
Overlap graph

Below: overlap graph, where an overlap is a suffix/prefix match of at least 3 characters

A vertex is a read, a directed edge is an overlap between suffix of source and prefix of sink

Vertices (reads): { *a*: CTCTAGGCC, *b*: GCCCTCAAT, *c*: CAATTTTT }

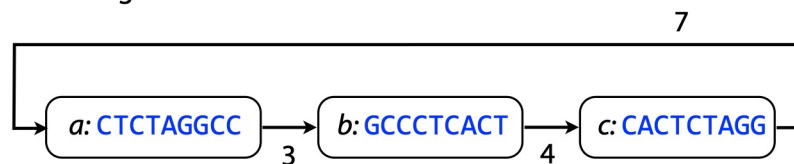
Edges (overlaps): { (*a*, *b*), (*b*, *c*) }



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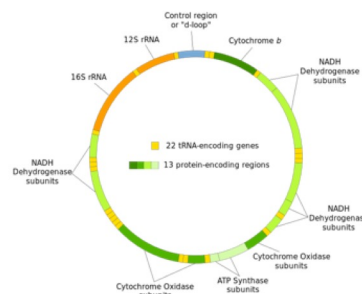
Overlap graph

Overlap graph could contain cycles. A cycle is a path beginning and ending at the same vertex.



These happen when the DNA string itself is circular. E.g. bacterial genomes are often circular; mitochondrial DNA is circular.

Cycles could also be due to *repetitive* DNA, as we'll see

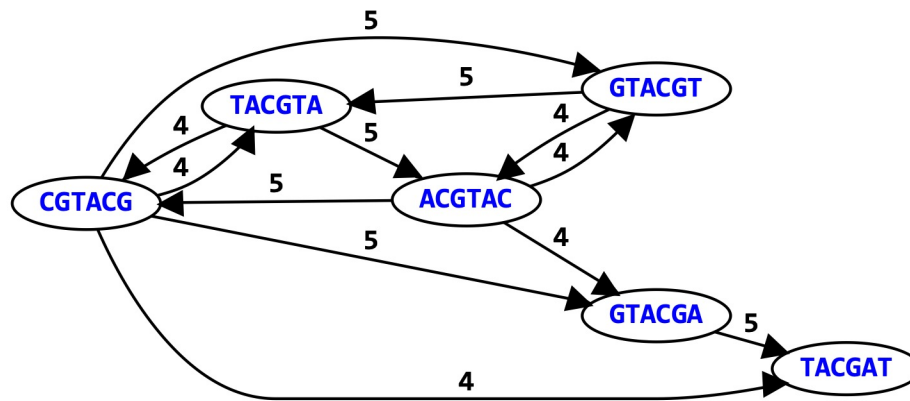


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Overlap graph Putting weight on the overlap

Nodes: all 6-mers from **GTACGTACGAT**

Edges: overlaps of length ≥ 4

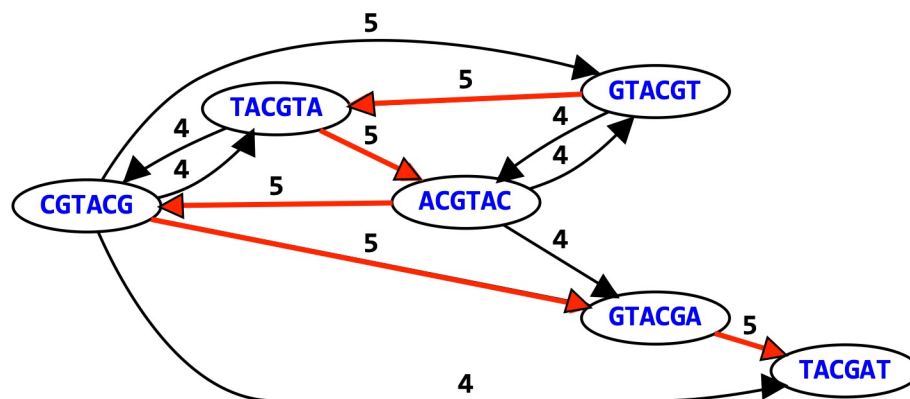


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Overlap graph

Nodes: all 6-mers from **GTACGTACGAT**

Edges: overlaps of length ≥ 4



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Directed multigraph

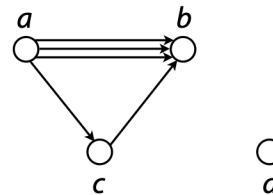
Directed **multigraph** $G(V, E)$ consists of set of vertices, V and **multiset** of directed edges, E

Otherwise, like a directed graph

Node's *indegree* = # incoming edges

Node's *outdegree* = # outgoing edges

De Bruijn graph is a directed multigraph



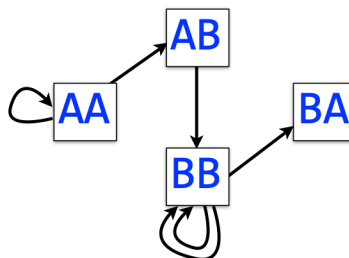
$$V = \{a, b, c, d\}$$

$$E = \{(a, b), (a, b), (a, b), (a, c), (c, b)\}$$

└──────── Repeated ─────────┘

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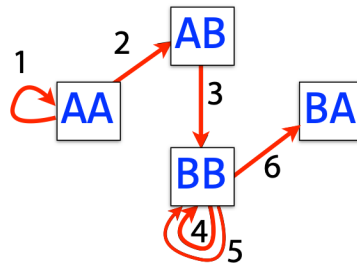
De Bruijn graph



Walk crossing each edge exactly once gives a reconstruction of the genome

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De Bruijn graph



AAABBBBA

Walk crossing each edge exactly once gives a reconstruction of the genome. This is an *Eulerian walk*.

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Eulerian walk definitions and statements

Node is *balanced* if indegree equals outdegree

Node is *semi-balanced* if indegree differs from outdegree by 1

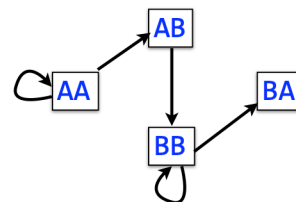
Graph is *connected* if each node can be reached by some other node

Eulerian walk visits each edge exactly once

Not all graphs have Eulerian walks. Graphs that do are *Eulerian*.
(For simplicity, we won't distinguish Eulerian from semi-Eulerian.)

A directed, connected graph is Eulerian if and only if it has at most 2 semi-balanced nodes and all other nodes are balanced

Jones and Pevzner section 8.8



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Third law: repeats make assemblies difficult

De Bruijn graph

No: graph can have multiple Eulerian walks, only one of which corresponds to original superstring

Right: graph for **ZABCDABEFABY**, $k = 3$

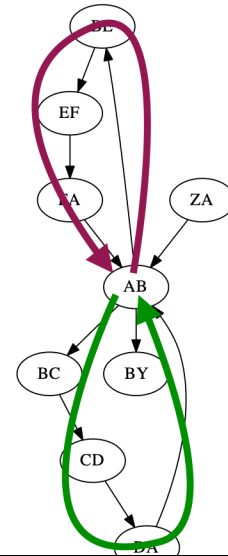
Alternative Eulerian walks:

ZA → **AB** → **BE** → **EF** → **FA** → **AB** → **BC** → **CD** → **DA** → **AB** → **BY**

ZA → **AB** → **BC** → **CD** → **DA** → **AB** → **BE** → **EF** → **FA** → **AB** → **BY**

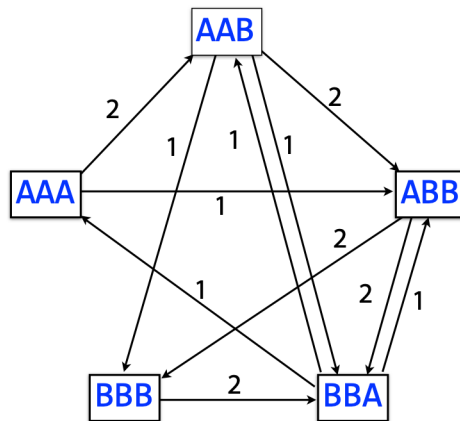
These correspond to two edge-disjoint directed cycles joined by node **AB**

AB is a repeat: **ZABCDABEFABY**



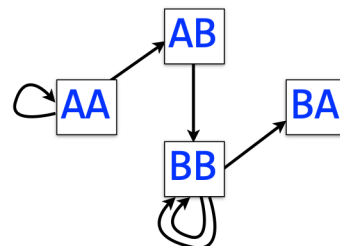
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Overlap graph



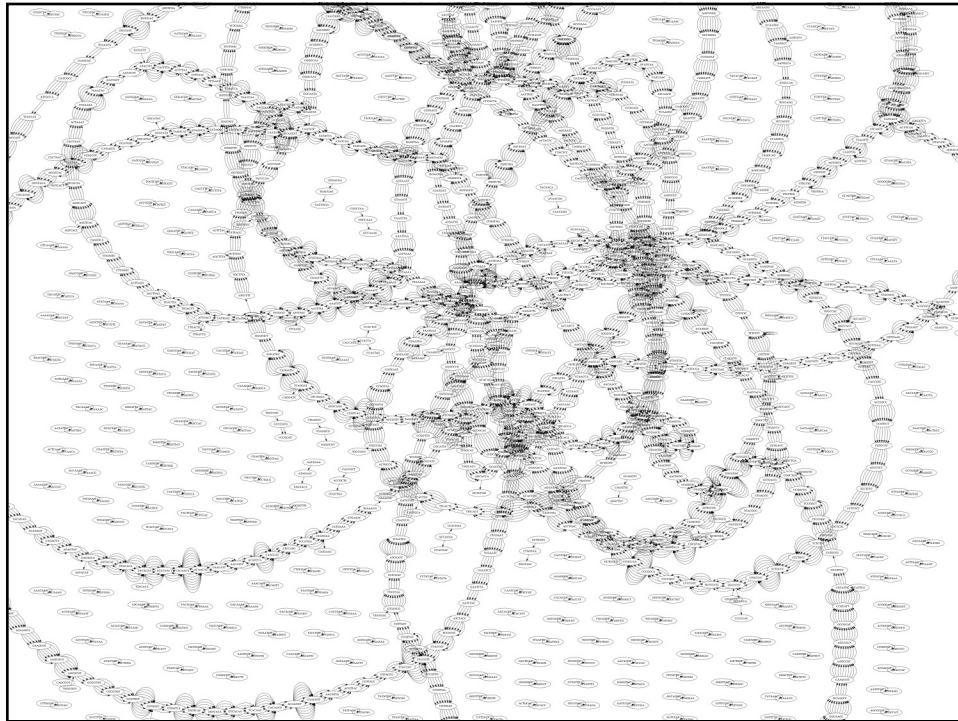
Overlap-Layout-Consensus
(OLC) assembly

De Bruijn graph

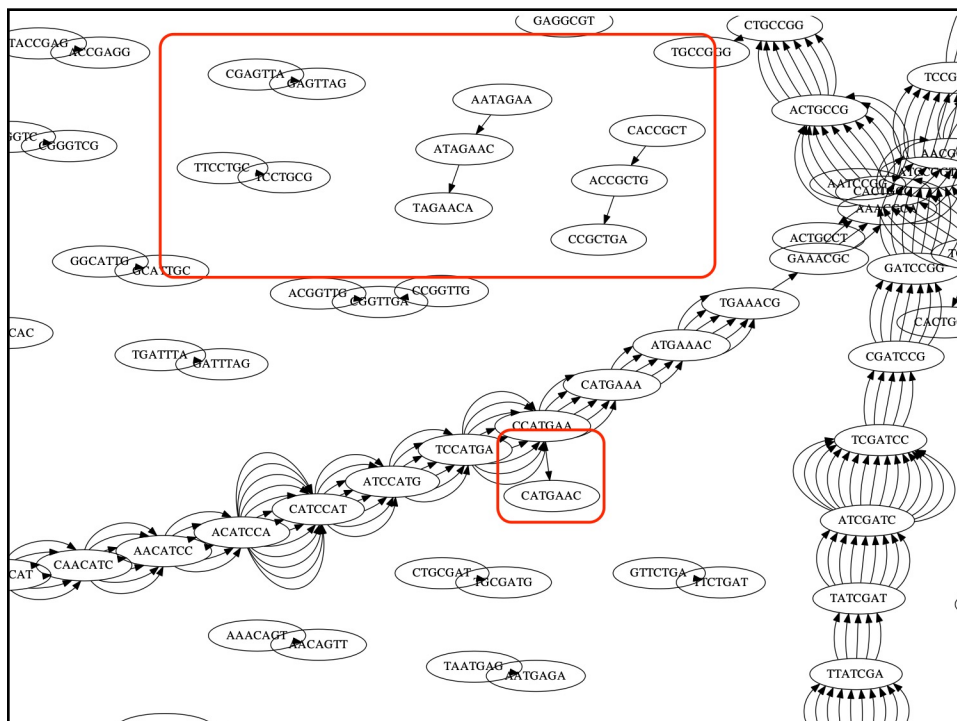


De Bruijn Graph based
(DBG) assembly

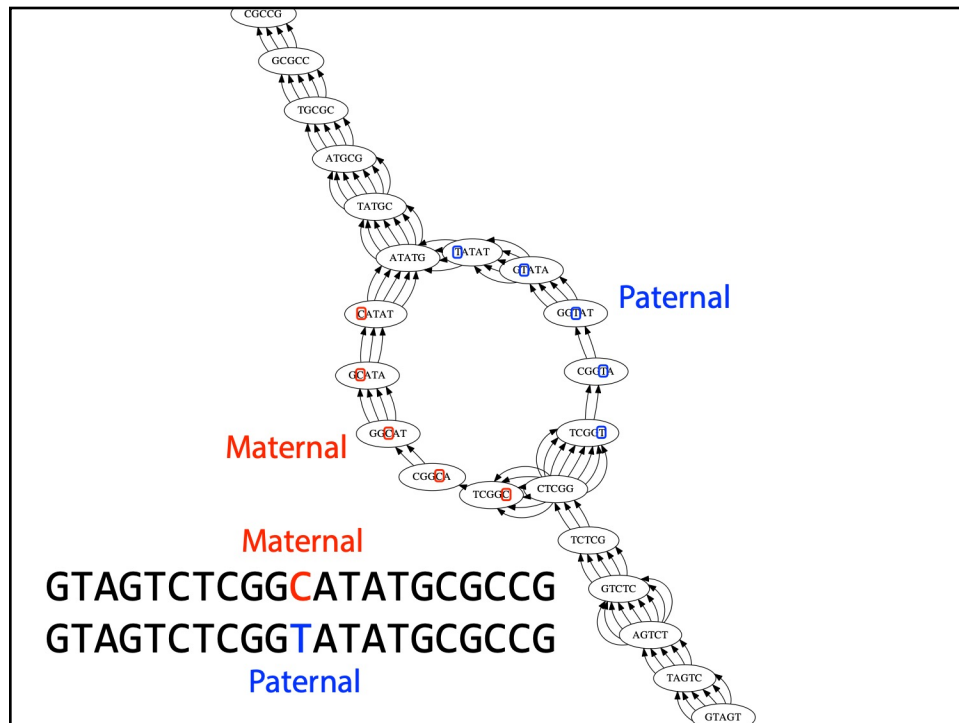
64



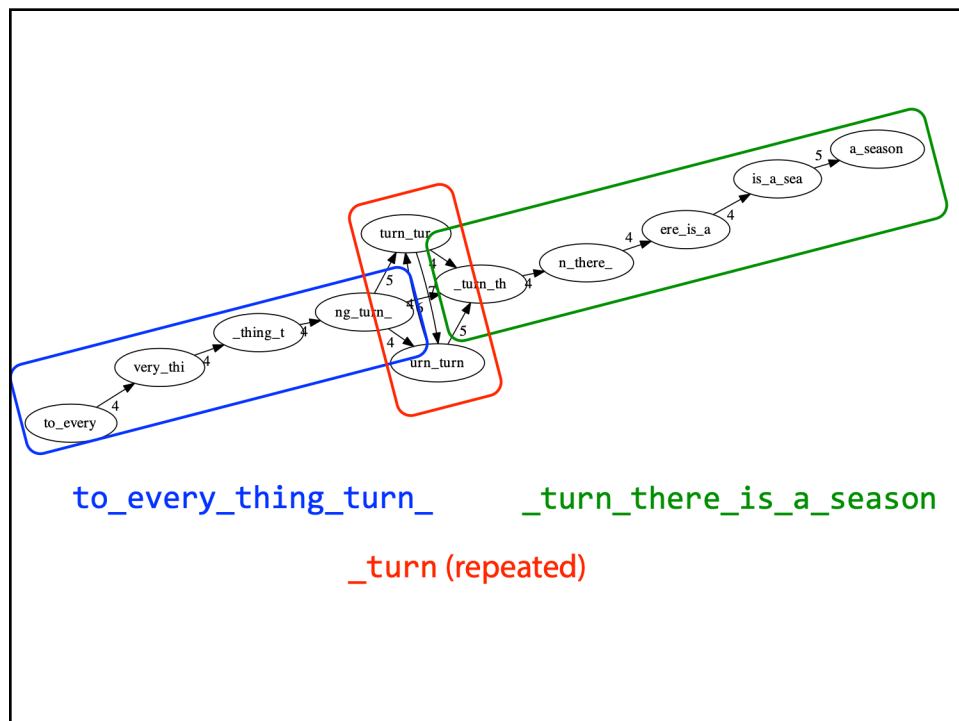
65



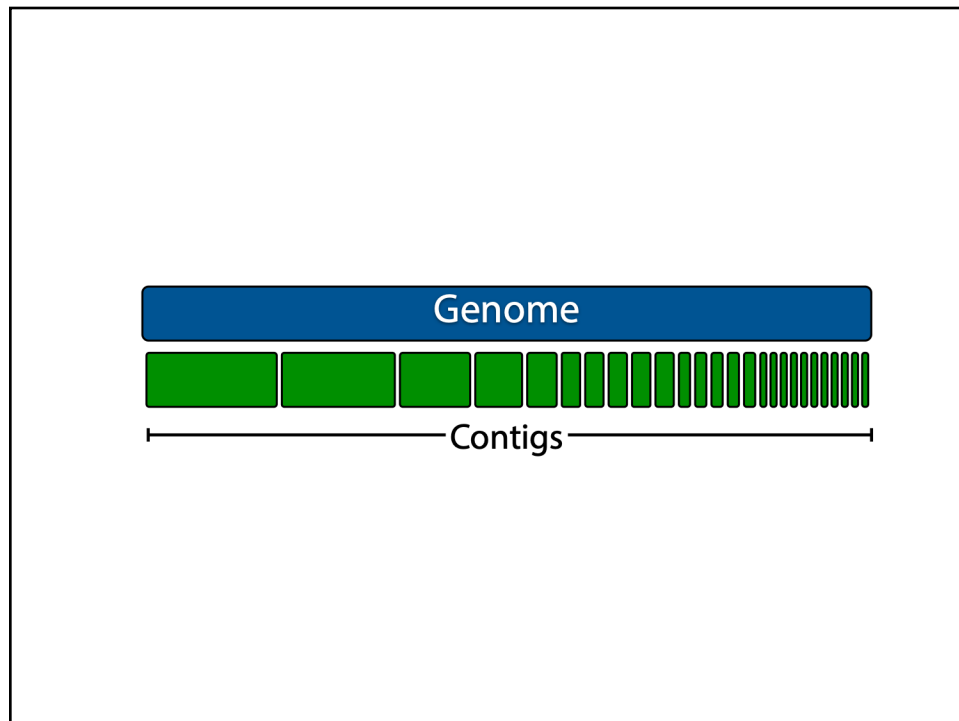
66



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N50 size

Def: 50% of the genome is in contigs as large as the N50 value

Example: 1 Mbp genome

50%



N50 size = 30 kbp

$(300k + 100k + 45k + 45k + 30k = 520k \geq 500kbp)$

Note:

N50 values are only meaningful to compare when base genome size is the same in all cases

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